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Studio Session	4

Part A: URL of the Visualisation

https://layela3.github.io/Data-Visualisation/

Part B: Domain Description

Describe your domain chosen for DV II

Australian Government Spending. Investigate Australian government spending, comparing expenditure levels, categories and changes over time. Explore international comparisons, spending categories and changes over time.

Describe the Why and Who aspects of your DV II

Who- The average Australian resident and taxpayer interested to know how public money is allocated.

Why- Government spending affects everyone living in Australia and it is sometimes difficult to understand where the money goes. The visualisation explores where expenditure goes, it moves from broad comparisons to state-level breakdowns and dissects the large spending on healthcare. This makes complex spending patterns easier to interpret.

A range of visualisations were used to answer some of the following questions: “is Australia a big spender”, “how does Australia compare with other countries”, “where does all the money go”, “how much have spending trends changed over time.”

Chart 1: Government revenue vs expenses. A line chart clearly shows changes over time; it is easy to identify when spending exceeds income and how the gap changes with time.
Chart 2: International spending comparison. A bar chart allows clear and direct comparison between Australia and other countries, making the different levels easy to notice.
Chart 3: Spending across states & territories. A proportional symbol map against the different state sizes at face value allows the reader to compare the scale of spending across

the states. This visualisation is purposefully not meant to be very clear as it follows the storyline.
Chart 4: State spending breakdown. A heatmap is very easy to read and compare spending allocations across states and categories. The darker colours emphasising more money spent in a category is also pragmatic.
Chart 5 & 7: The lollipop charts are for easy comparison across the states in different categories.
Chart 6: Hospital locations. The dot map was used to reveal spatial patterns and the scarcity in some remote Australian areas. It is very easy to spot the concentrated places.
Chart 8: Health spending categories. Pie chart simply shows the distribution clearly and shows that even though there are a few sub-categories, the whole budget is going to health.
Chart 9: Spending changes over a decade. The dot plot compares spending between two years and clearly shows the magnitude of change over time.
Chart 10: Australia's spending ranking over time. Bump chart clearly shows the change in budget and rank over time. It is easier to map the changes of spending through the years compared to other country changes.
Chart 11: Global health spending. Choropleth map allows the reader to see geographic patterns whilst comparing Australia with other countries. It is also pragmatic since the darker colours represent higher spending per person.

Describe the "What" aspect of your DV II

Chart 1: Government revenue vs expenses (quantitative attribute) are displayed across multiple years (temporal attribute) in a line chart.
Chart 2: Government spending (quantitative) is compared across countries (categorical) in a bar chart.
Chart 3: Australian states (categorical) are represented geographically (spatial) with proportional symbols of the total government expenditure (quantitative) on the map.
Chart 4: The heatmap has states and spending category (quantitative attributes) and displays the percentage of total expenditure (quantitative).
Chart 5: Government expenditure (quantitative) across spending categories (categorical) is represented in the lollipop chart.
Chart 6: The dot map shows hospital locations (latitude and longitude – spatial attributes) plotted on the Australian map (spatial)
Chart 7: Health spending per person (quantitative) is compared across Australian states (categorical) in the lollipop chart
Chart 8: The pie chart shows the health service categories (categorical) and their share of the total health expenditure (quantitative).
Chart 9: Connected dot plot compares government expenditure (quantitative) across the years (temporal) for each spending category (categorical).

Chart 10: The bump chart tracks Australia's position relative to other countries (categorical) across multiple years (temporal)
Chart 11: The choropleth has Countries (spatial) shaded according to their government health expenditure per person (quantitative).

Describe the "How" aspect of your DV II

Chart 1: Marks- lines and points.
Channels- horizontal position showing the year. Vertical position shows the revenue or expenses. Colour hue distinguishes revenue(blue) from expenses(teal).

Chart 2: Marks- lines(2D)
Channels- length used to show the spending per capita. Colour hue highlights Australia from other countries. Horizontal position shows the spendings per capita

Chart 3: Marks- points(2D)
Channels- Geographic position places each symbol at its corresponding state. Size (area) represents the expenditure with larger circles indicating higher spending.

Chart 4: Marks- point (2D)
Channels- Horizontal position shows spending categories. Vertical position shows states. Colour luminance represents the percentage of total expenditure.

Chart 5: Marks- line and point
Channels- Horizontal position shows the spendings. Vertical position identifies Australian states. Length shows distance between axis ad spending value.

Chart 6: Marks- points
Channels- Geographic position shows latitude and longitude for each hospital. Colour hue distinguishes public and private hospitals.

Chart 7: Marks- lines and points
Channels-Horizonal position shows spending per person. Vertical position shows the Australian state. Length shows distance between the axis and spending value per person.

Chart 8: Marks- area
Channels- Size (area) and Angle shows each category's proportion of total health spending. Colour hue distinguishes health services.

Chart 9: Marks- points and lines

Channels- Horizontal position shows the government spending. Vertical position shows the spending categories. Colour hue distinguishes 2015 and 2025. Length indicates magnitude of change over time.

Chart 10: Marks- lines and points

Channels- Horizontal position shows the year. Vertical position shows the international rankings. Colour hue distinguishes Australia from other countries.

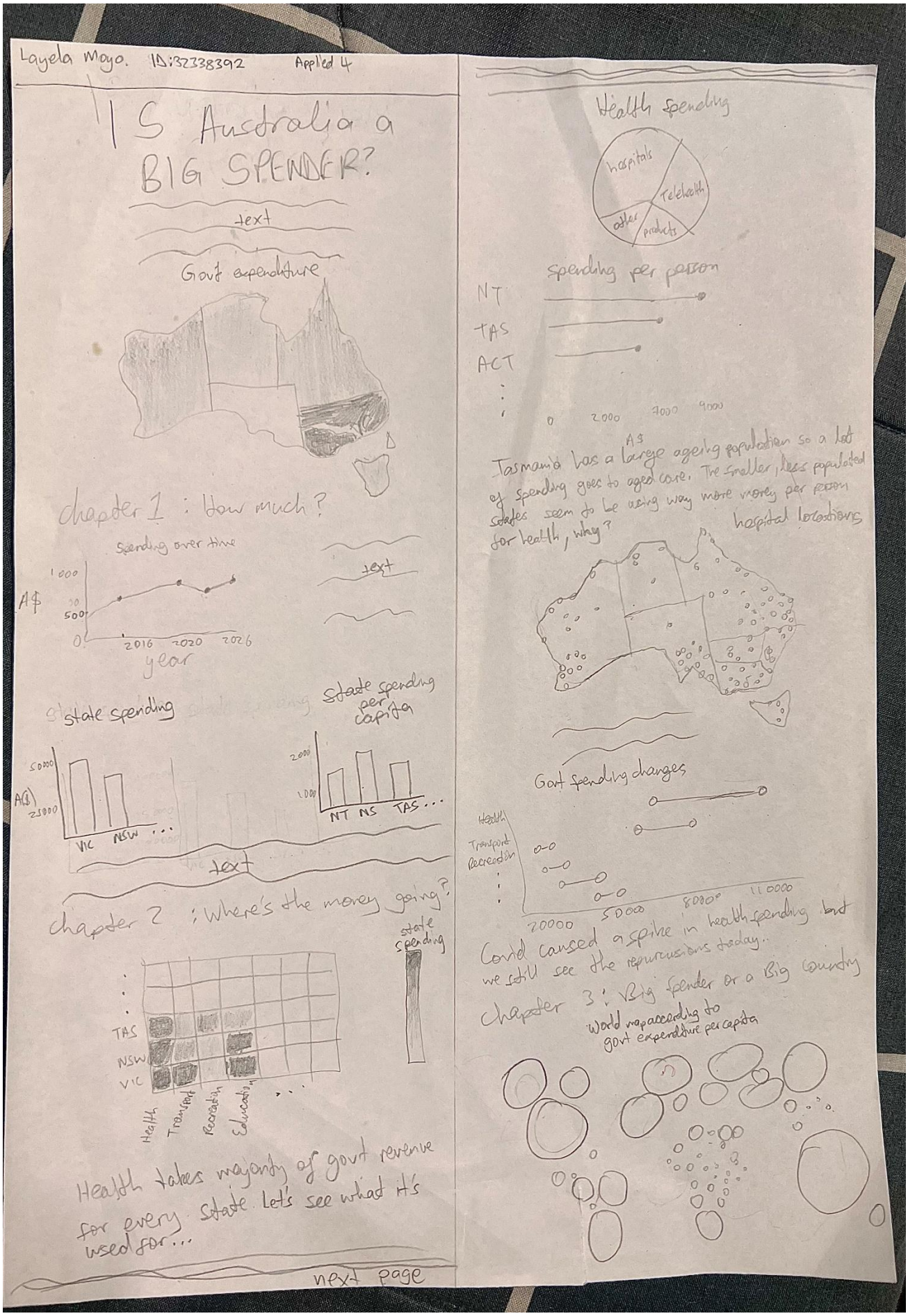
Chart 11: Marks- area

Channels- Geographic position and shape shows the different countries. Colour luminance shows health expenditure per person.

Part C: Sketch

https://github.com/layela3/Data-Visualisation/blob/main/docs/Layela%20sketch.pdf

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Layela Moyo. ID:32338392 Applied 4
IS Australia a BIG SPENDER?
text
Govt expenditure
chapter 1 : How much?
Spending over time
A$
1000
500
0
2016
2020
2026
year
text
state spending
A($)
50000
25000
VIC
NSW
...
state spending per capita
2000
1000
NT
NS
TAS...
text
chapter 2 : Where's the money going?
state spending
TAS
NSW
VIC
Health
Transport
Recreation
Education
Health takes majority of govt revenue for every State. Let's see what it's used for...
next page
Health spending
hospitals
Telehealth
other
products
spending per person
NT
TAS
ACT
0
2000
A$
Tasmania has a large ageing population so a lot of spending goes to aged care. The smaller, less populated states seem to be using way more money per person for health, why?
hospital locations
Govt spending changes
Health
Transport
Recreation
20000
50000
80000
110000
Covid caused a spike in health spending but we still see the repurcussions today.
chapter 3 : Big spender or a Big country
world map according to govt expenditure per capita

Part D: Full-size screenshot of your DV II

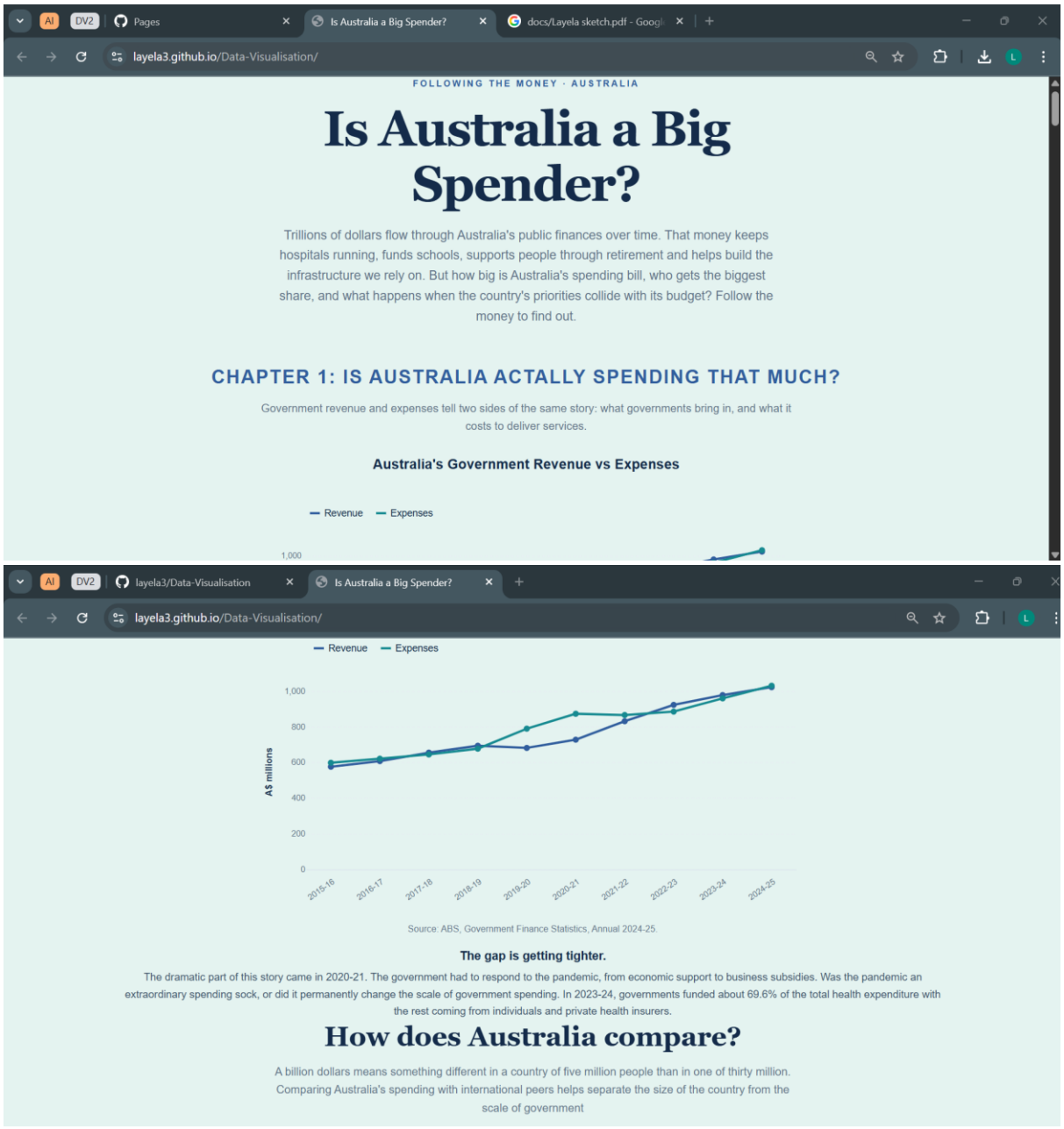


Chart 1: Line chart

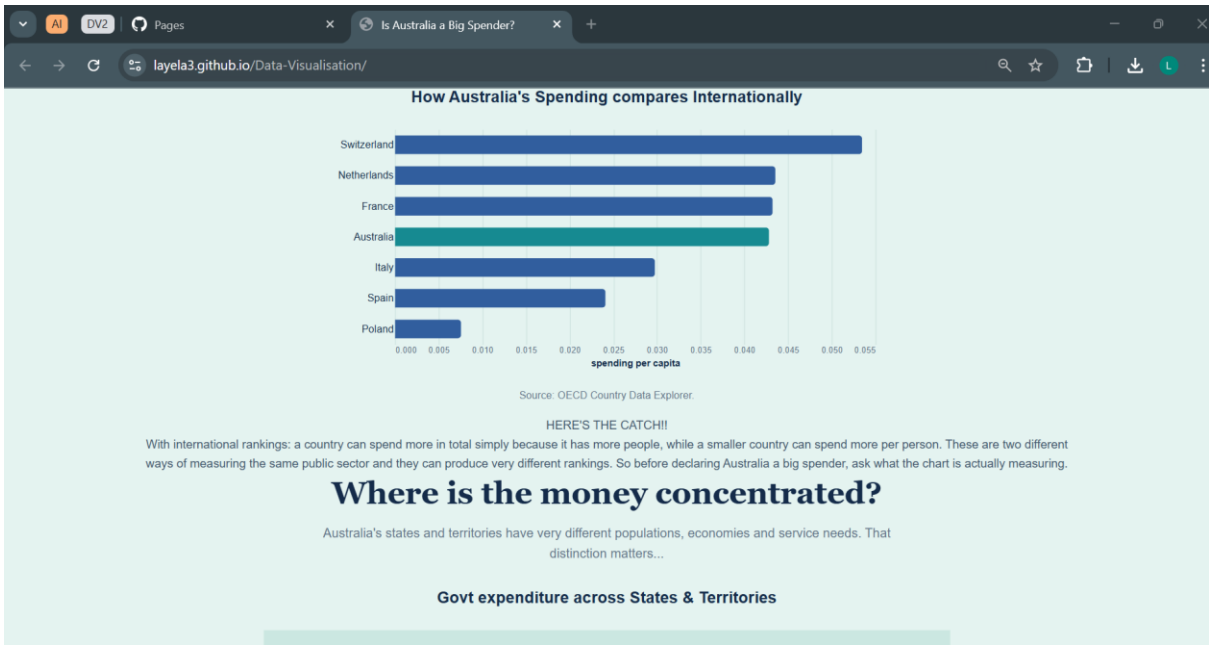


Chart 2: bar chart

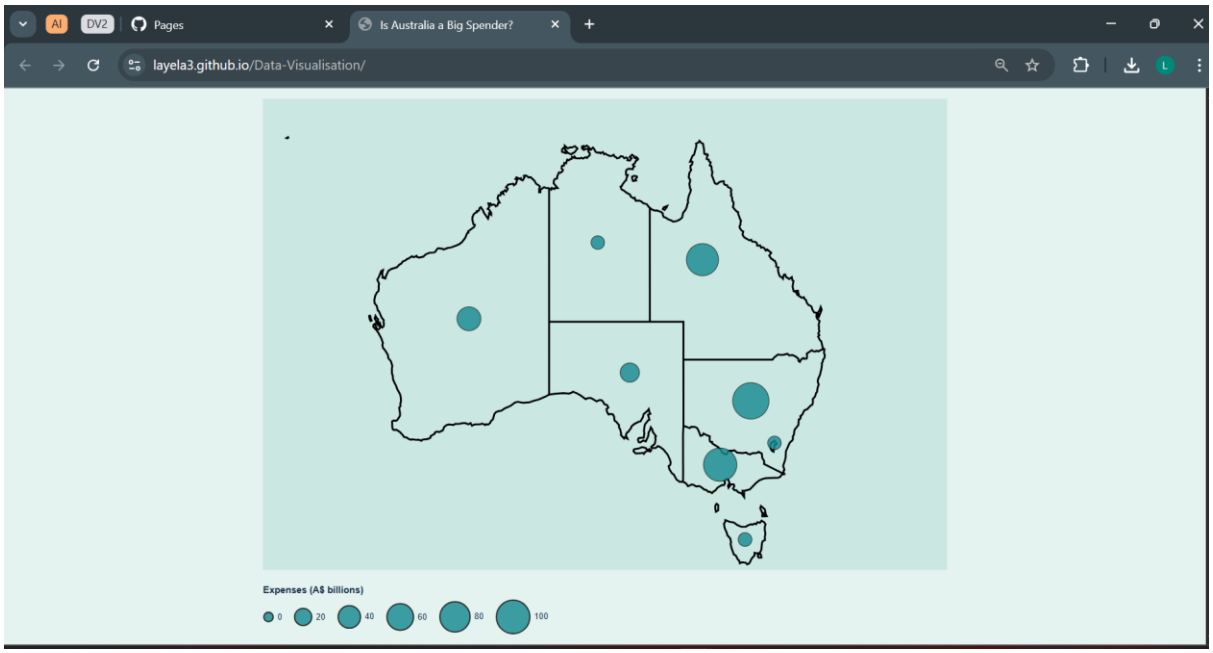


Chart 3: proportional symbol map

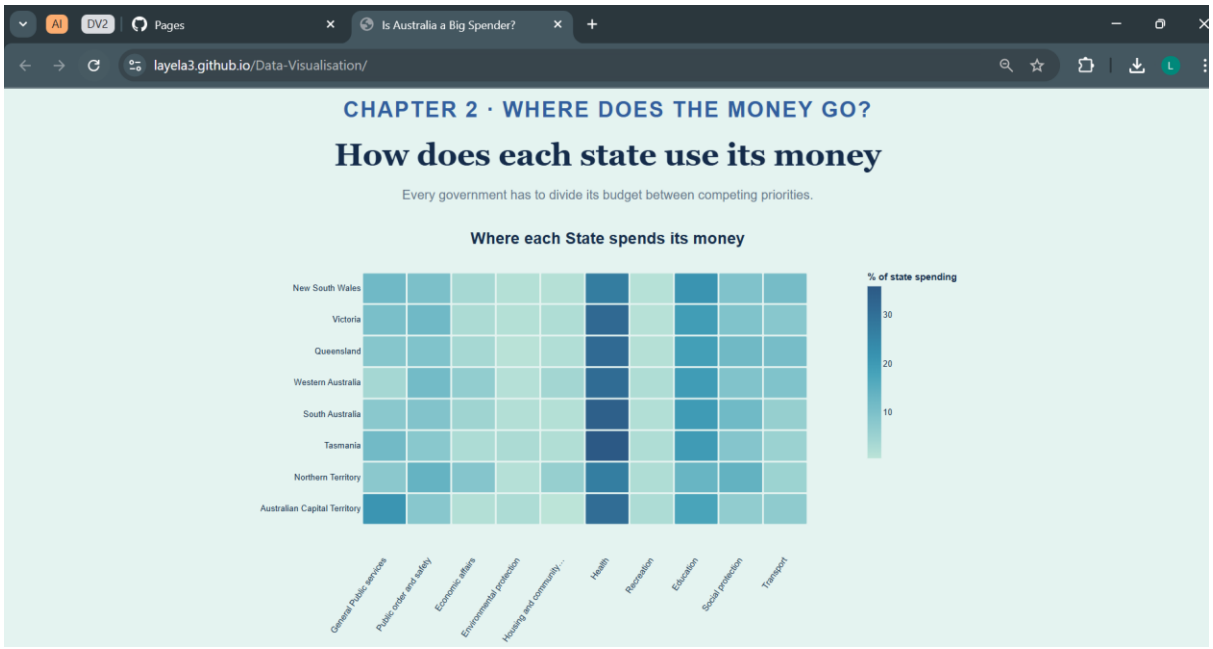


Chart 4: heatmap



Chart 5: lollipop chart

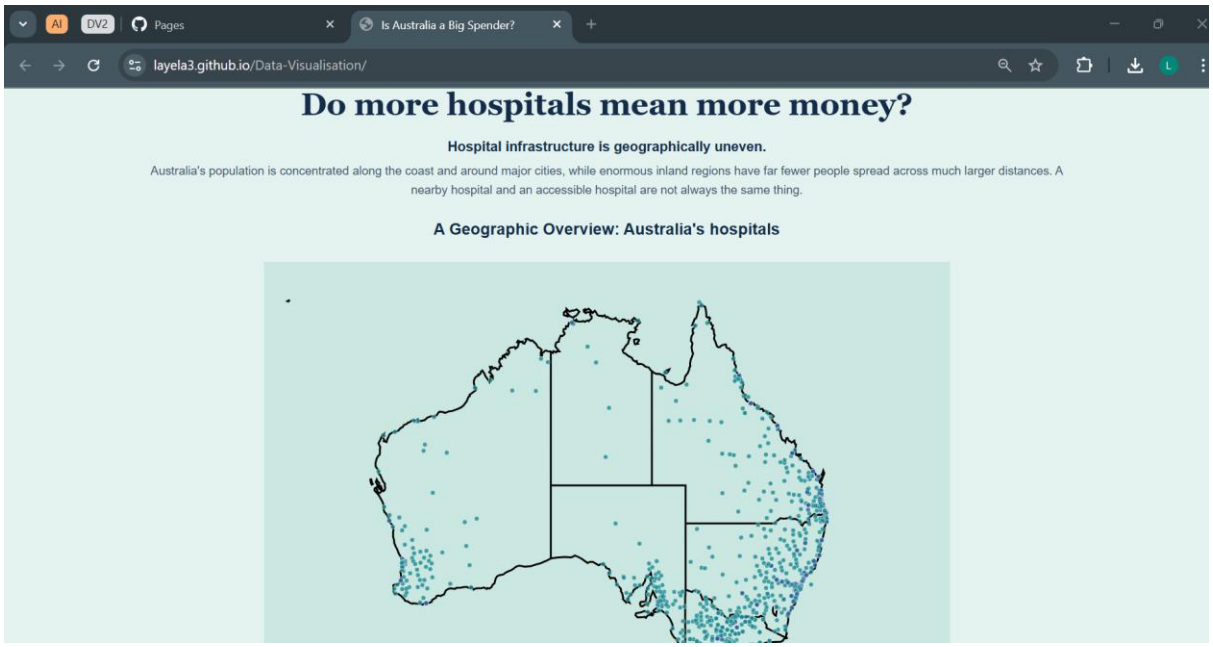
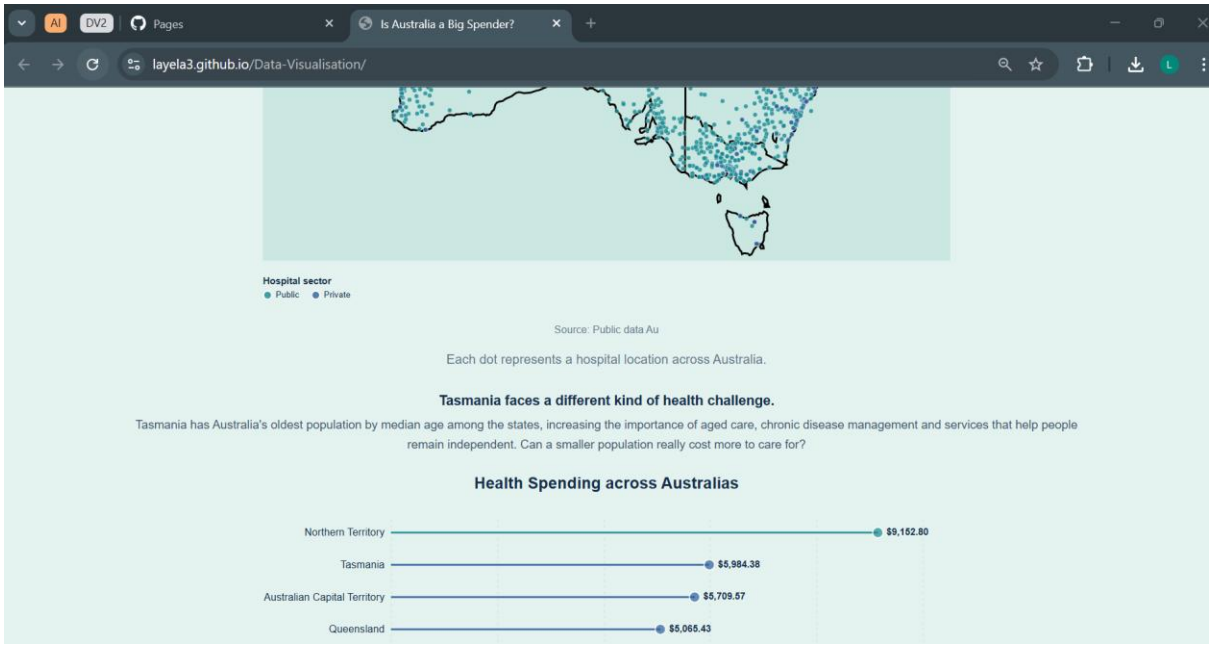


Chart 6: dot map



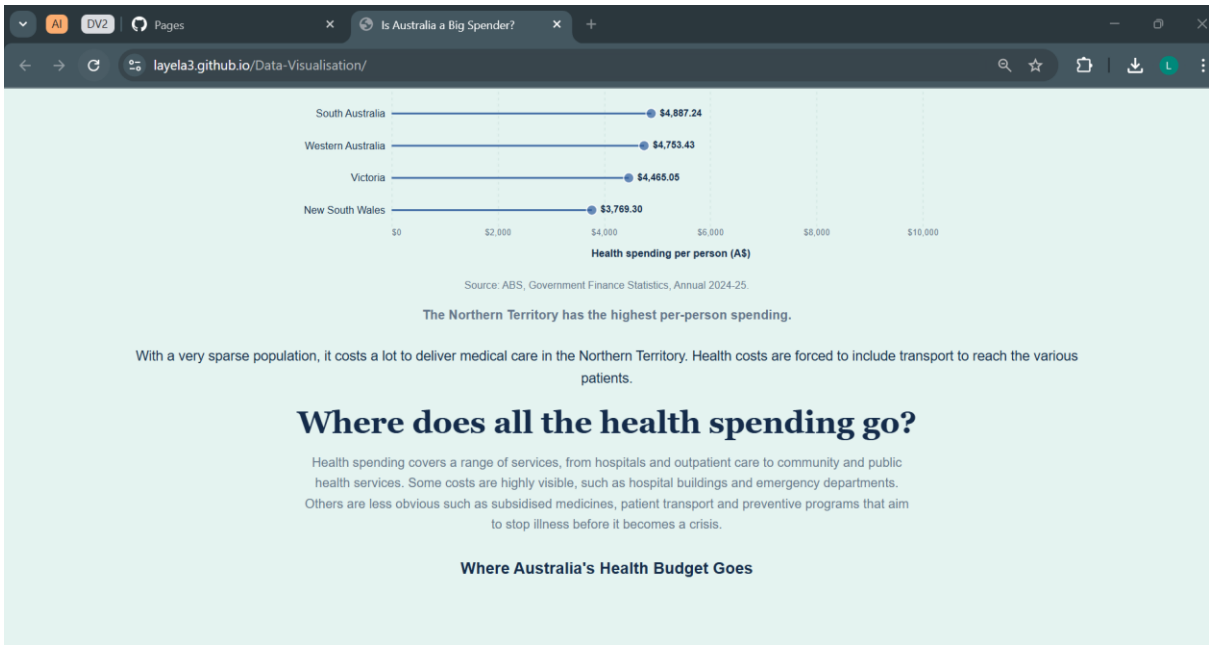


Chart 7: lollipop chart

chart

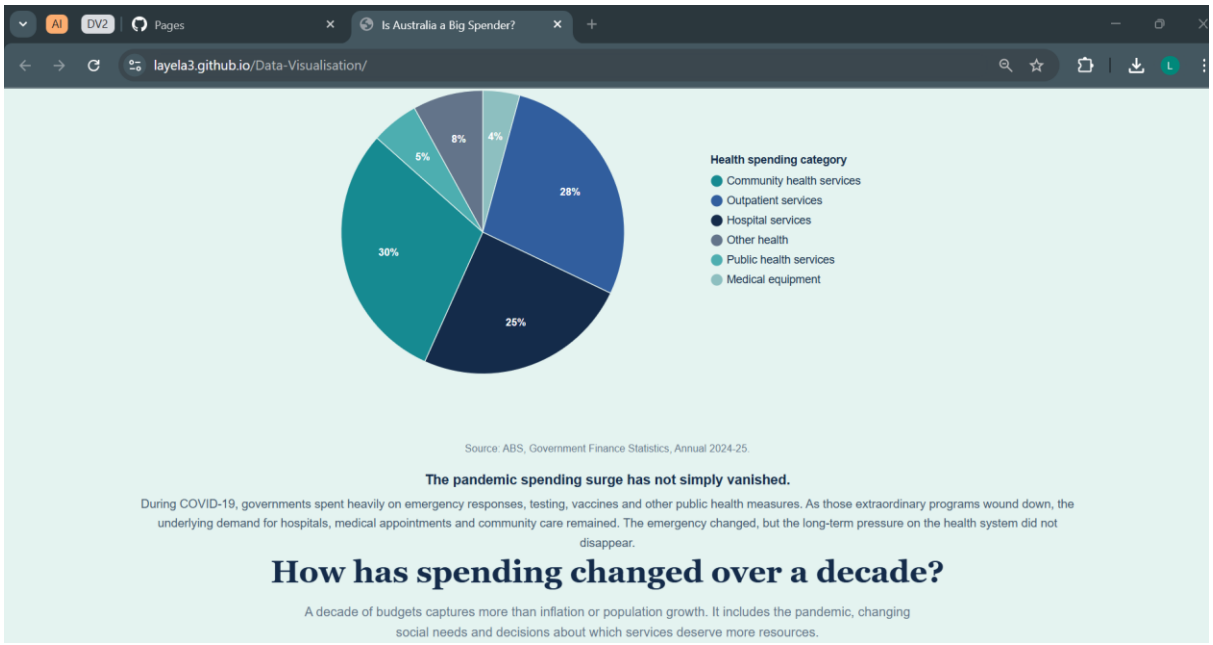


Chart 8: pie chart

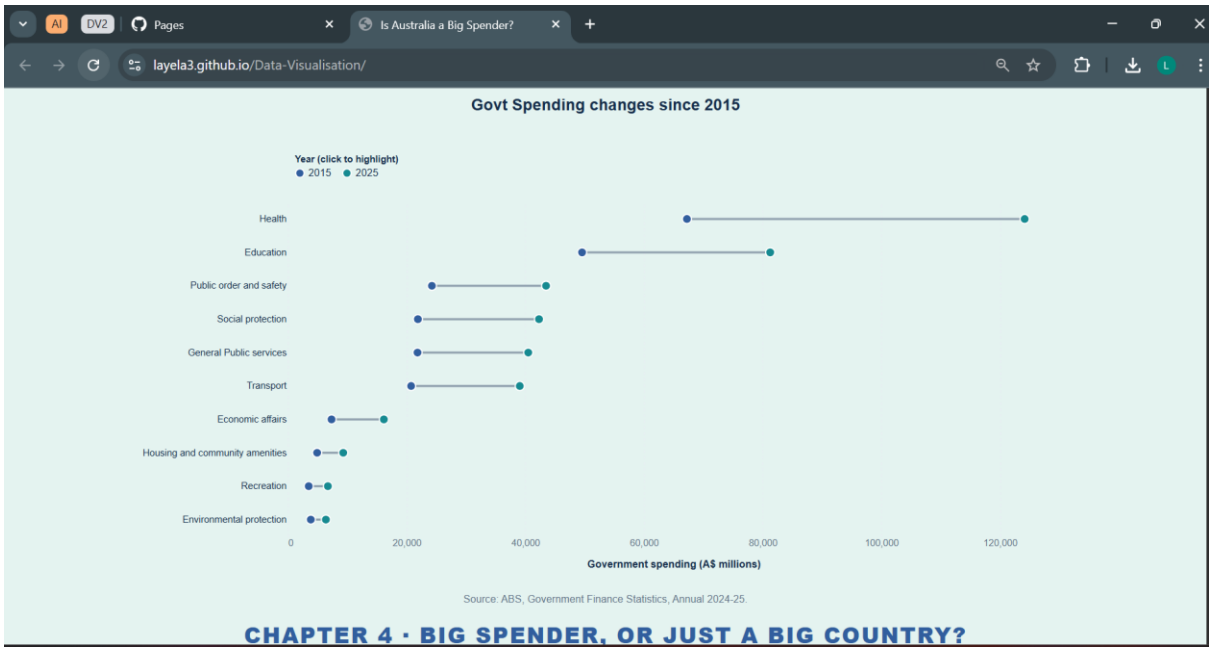


Chart 9: dot plot

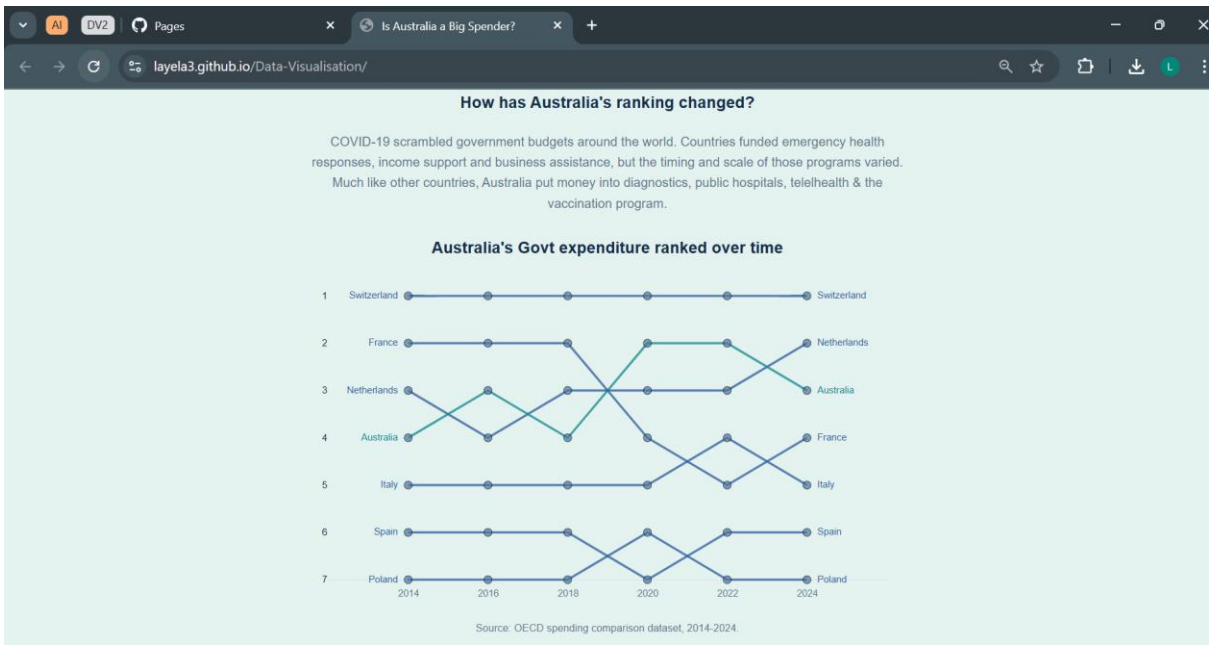
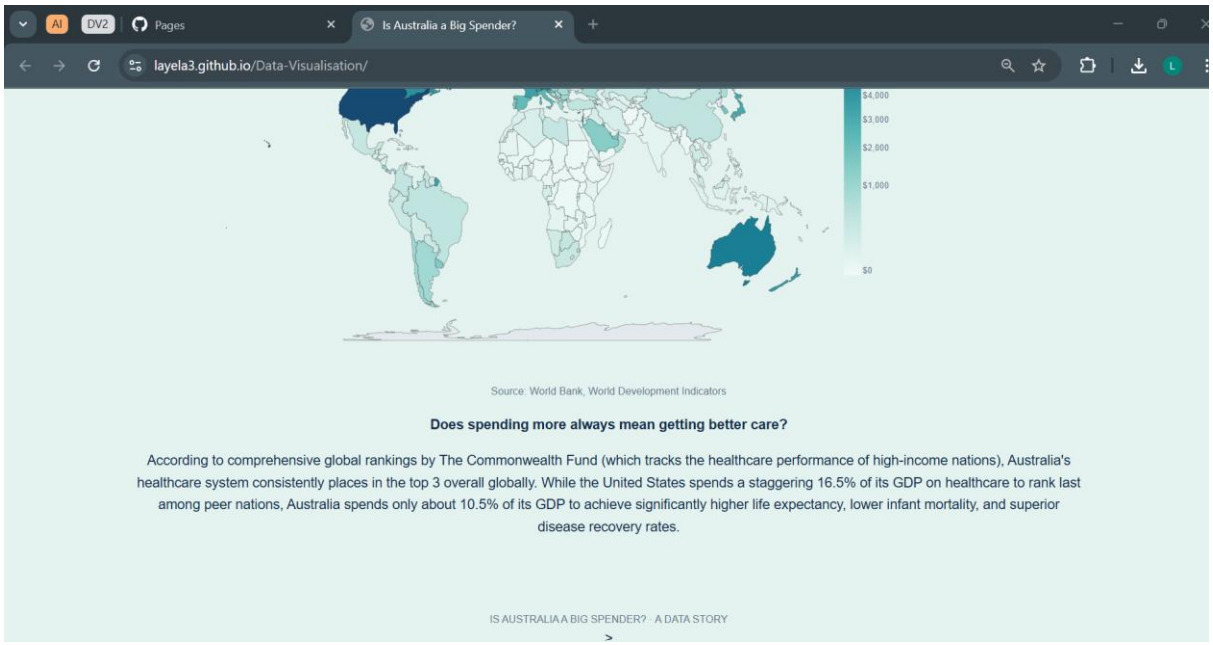


Chart 10: bump chart



Chart 11: choropleth map



AI declaration: I used ChatGPT to fix the grammar of the text when I was answering what, why, how, why questions.